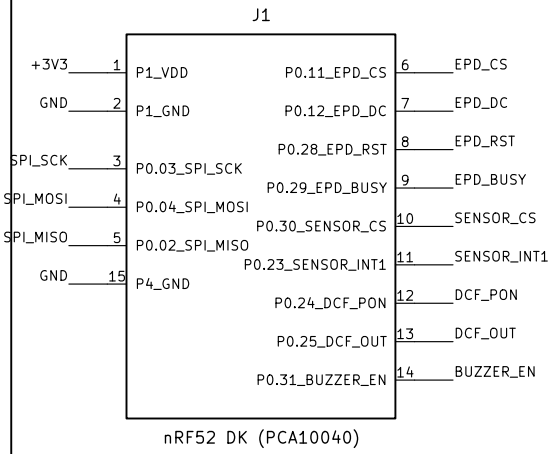
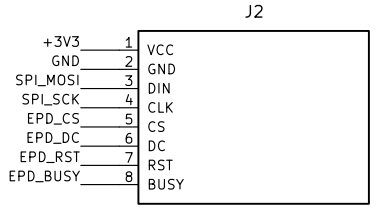
BLOCK 1: HOST CONTROLLER INTERFACE (nRF52 DK / PCA10040)

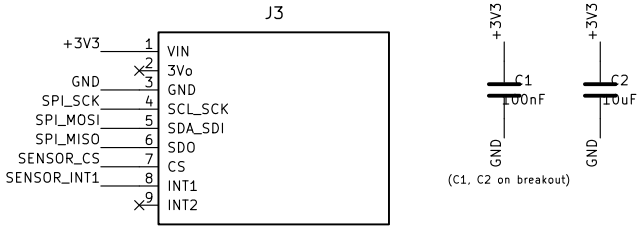


BLOCK 2: DISPLAY INTERFACE (Waveshare 2.13in HAT)

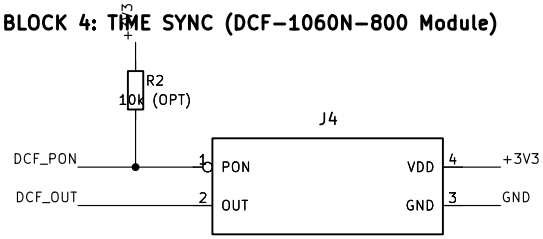


Verify BS1 = 0 (4-wire SPI) on HAT rear switch

BLOCK 3: MOTION SENSOR (Adafruit LIS3DH #2809)



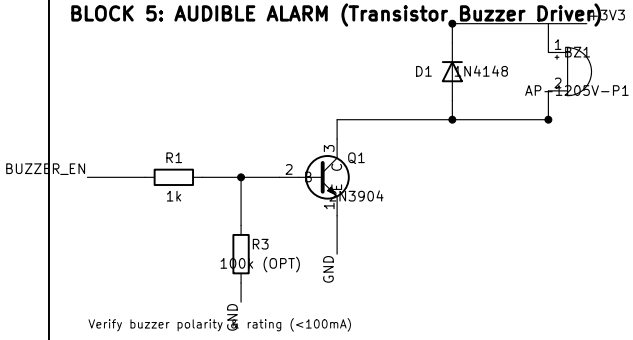
BLOCK 4: TIME SYNC (DCF-1060N-800 Module)



PON = 0V: Enabled (~68 uA) | PON = 3.3V: Disabled (0.0 uA)

MEASURE OUT HIGH LEVEL BEFORE CONNECTING DK P0.25!

BLOCK 5: AUDIBLE ALARM (Transistor Buzzer Driver)



Verify buzzer polarity & rating (<100mA)

PROTOTYPE SAFETY & INTEGRATION RULES

1. POWER: Feed the 3V3 rail only from DK header P1 pin VDD (USB powered). Never use 5V, External supply, VDD_nRF or P22. NEVER connect KORAD bench supply and the 3V3 rail together!
2. SAFE CONTROL STATES: Set DCF_PON HIGH (disabled, 0.0uA). Set EPD_CS and SENSOR_CS HIGH (unselected). Set BUZZER_EN LOW (off).
3. DCF-77 VERIFICATION: Measure DCF OUT high-level voltage before connecting to DK P0.25. Orient antenna perpendicular to Frankfurt.
4. E-PAPER CONFIGURATION: Verify switch BS1 = 0 on back of Waveshare HAT for 4-wire SPI mode before powering display rail.
5. BUZZER DRIVER: Verify 2N3904 TO-92 pinout (1=Emitter, 2=Base, 3=Collector). Diode D1 cathode connects to +3V3 to clamp flyback spikes.
6. STAGED BRING-UP: 1) DK Blinky -> 2) E-Paper Demo -> 3) LIS3DH WHO_AM_I -> 4) DCF Standalone -> 5) DCF Sync -> 6) Buzzer Alarm.

nRF52 DK (PCA10040) + E-Paper + LIS3DH + DCF-77 + Buzzer Driver

Bachelor Thesis (APD)

Sheet: /

File: breadboard-prototype.kicad_sch

Title: Ultra-Low-Power Bedside Clock — Breadboard Prototype

Size: A4	Date: 2026-09-13	**Rev: v1.1**
KiCad E.D.A. 10.0.4		Id: 1/1